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The Association between Myopia and Anxiety: A Bidirectional 2-Sample Mendelian Randomization Study

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Objective: To investigate whether there is a causal relationship between generalized anxiety disorder (GAD) and myopia.

Design: A bidirectional 2-sample Mendelian randomization (MR) analysis.

Subjects: For the forward MR, 11 single nucleotide polymorphisms (SNPs) associated with GAD were obtained from the FinnGen Research (5280 cases and 368 054 controls; European). For reverse MR, 253 SNPs associated with mean spherical equivalent (MSE) were derived from the UK Biobank (95 619 participants; European).

Methods: Summary-level data from the 2 genome-wide association studies were used to conduct bidirectional MR analyses. Four MR methodologies—inverse variance weighted (IVW), MR-Egger regression, weighted median, and weighted mode—were employed, with IVW serving as the primary analysis. Sensitivity analyses, including MR-Egger intercept and heterogeneity Q test, were performed.

Main Outcome Measures: The outcomes of the forward and reverse MR analyses were MSE and GAD, respectively.

Results: This study demonstrated a significant causal effect of GAD and myopia, suggesting individuals with GAD are at an elevated risk of more myopic MSE (IVW: $\beta = -0.138$; 95% confidence interval [CI] −0.221 to −0.056, $P = 0.001$). Conversely, we did not find evidence to support a causal effect of myopia on GAD (IVW: $\beta = -0.017$; 95% CI −0. 051 to −0.016, $P = 0.308$). Sensitivity analyses revealed no evidence of genetic confounding to bias the estimate of causal effect.

Conclusions: Our study demonstrates a significant causal relationship between GAD and a more myopic MSE, though the effect size was relatively small. These findings could be taken into consideration while implementing mental health interventions in children.

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The global prevalence of myopia is rising significantly as a public health concern. It is projected that over half of the world's population will be affected by 2050, with East and Southeast Asia most affected.[1] The prevalence of myopia is also on the rise in Europe,[2,3] where up to 47.2% of young adults aged 25 to 29 years are affected.[4] Myopia can lead to severe complications such as retinal detachment, macular degeneration, and glaucoma, which are major causes of vision loss and blindness.[5] The etiology of myopia is complex and multifactorial, encompassing a diverse interplay of genetic, environmental, and lifestyle factors that are still inadequately understood.[6,7] Crucial risk factors identified include higher levels of education, extensive near work, and limited outdoor activities.[8–10]

Anxiety disorders also represent a significant global public health issue, with an estimated 301.39 million prevalent cases and 45.82 million incident cases reported worldwide in 2019.[11,12] Western Europe exhibited an age-standardized incidence rate of 7.91 per 1000 populations in 2019, surpassing the global average of 5.85 per 1000 populations.[11,12] Among European youth, anxiety remains the most common mental health condition, affecting 7.9% of children and adolescents.[13]

Recent research has highlighted a significant relationship between myopia and mental health.[14–17] Individuals with myopia are reported to be at a higher risk of having anxiety, including generalized anxiety disorder (GAD).[14,16,18,19] On the other hand, patients with chronic stress were reported to have an increased ocular axial length elongation rate.[20] However, existing evidence mostly came from cross-sectional studies, and whether there is a causal relationship between anxiety and myopia remains unknown. The concurrent rising prevalence of myopia and anxiety underscores the urgent need to explore and clarify the relationship between these 2 conditions.[21]

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Understanding whether anxiety contributes to myopia or vice versa could have significant clinical implications, offering opportunities for early intervention and integrated management strategies.

Mendelian randomization (MR) is a robust tool for inferring causality between modifiable exposures and health outcomes using genetic variants as instruments.[22–25] Taking advantage of Mendel laws of heredity, MR ensures random allele assignment in offspring, which helps reduce potential biases such as confounding and reverse causation, common issues in observational research.[26–29] In this study, a bidirectional 2-sample MR analysis using large-scale genome-wide association studies (GWAS) was employed to investigate the relationship between anxiety and myopia.

Methods

Overview of Study Design

A brief description of the study design is shown in Figure 1. The 3 assumptions of MR were followed to guarantee the reliability of results:[30] (1) There is a robust correlation between genetic variables and the exposure; (2) there is no correlation between the genetic variables and confounding factors; and (3) the genetic variables affect the outcome only through exposure.

GWAS Summary Statistics of Myopia and Anxiety

Two publicly available GWAS were involved in this MR study. Summary statistics of myopia, defined as noncycloplegic autorefraction–measured mean spherical equivalent (MSE), were obtained from the UK Biobank, including 95 619 individuals of European ancestry aged 40 to 69 years.[31] Though MSE was measured without cycloplegia, it is unlikely to bias the MSE since the accommodative amplitude is considerably lower in adults.[32–34] Genetic statistics for GAD were obtained from FinnGen Research[35] (https://r10.finngen.fi/pheno/F5 GAD), involving 5280 hospital discharge cases and 368 054 controls with a median age of first GAD onset at 38.23 years (men = 40.17, women = 37.51). Generalized anxiety disorder was identified based on hospitalization records using the *International Classification of Diseases*, Ninth Revision (300.02) and *International Classification of Diseases*, 10th Revision (F41.1) standards. According to these criteria, a diagnosis is made when individuals exhibit excessive fear or worry accompanied by ≥3 of the following symptoms: fatigue, restlessness, irritability, difficulty concentrating, muscle tension, and sleep disturbances.

This secondary analysis used deidentified data sourced from public databases. The original studies obtained ethical approvals from institutional review boards. Informed consent was secured during the primary data collection phase. Because no new data collection or participant interaction took place, additional ethical approval was deemed unnecessary according to the Declaration of Helsinki and the guidelines for anonymized data reuse.

Selection of Instrumental Variables

First, genome-wide significance of $P < 5 \times 10^{-6}$ was used to identify single nucleotide polymorphisms (SNPs) robustly associated with MSE or the presence of GAD. Second, to avoid the bias of linkage disequilibrium, instrumental variables (IVs) with linkage disequilibrium $r^2 < 0.001$ at a 10 000 kb window were removed to ensure the independence of SNPs. Third, an F-statistic was calculated to ascertain the robustness of each IV, which exceeded 10 and was considered sufficient to indicate the strength of IVs.[36] Summary statistics of exposure and outcome were harmonized to exclude palindromic and incompatible SNPs. The fitting SNPs would be chosen as the final IVs for the following MR analysis.

MR Analyses

We employed several analytical methods to estimate the bidirectional causal relationship between myopia and anxiety. Inverse variance weighted (IVW) serves as the primary approach, for it is the most efficient method for valid IVs, accounting for heterogeneity in causal estimates.[37] Other methods, including MR-Egger regression, weighted median, and weighted mode methods, were combined to further validate the robustness of the causal effect. Mendelian randomization-Egger regression assumes that more than half of IVs are being influenced by the horizontal pleiotropy; weighted median presumes that less than half of IVs are affected by horizontal pleiotropy; and weighted mode is used as a complementary analysis.[37] Estimates of these 3 methods should be directionally consistent with IVW to ensure pleiotropy does not bias IVW.

In all these analysis, $P < 0.05$ is considered significant.[37]

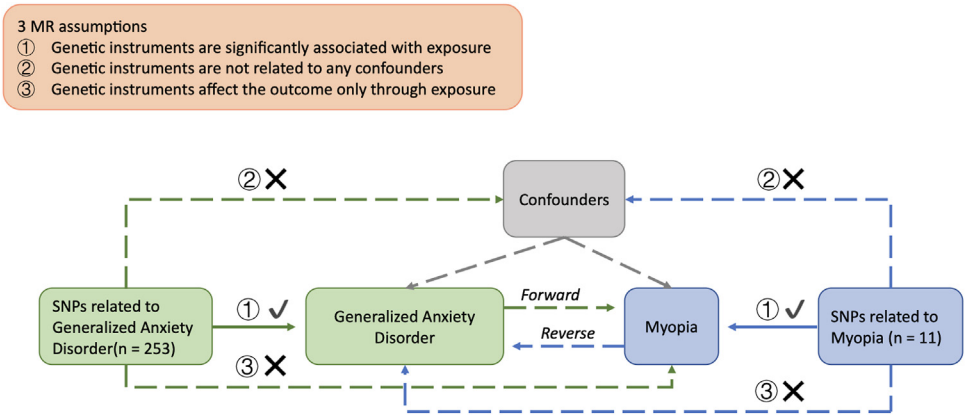


Figure 1. Overview of study design. The 3 assumptions of MR were followed to guarantee the reliability of results: (1) there is a robust correlation between genetic variables and the exposure; (2) there is no correlation between the genetic variables and confounding factors; and (3) the genetic variables affect the outcome only through exposure. MR = Mendelian randomization; SNPs = single nucleotide polymorphisms.

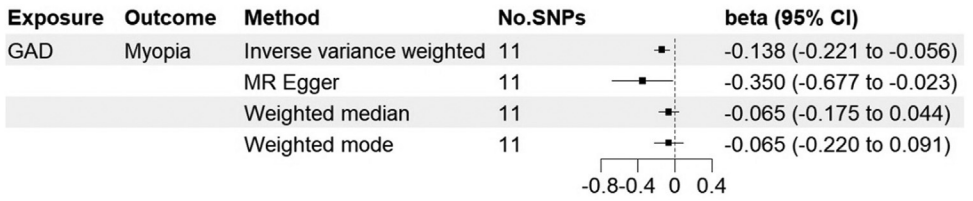


Figure 2. Four methods were employed to evaluate the association between GAD and myopia MSE. Forest plot displaying the causal effects of SNPs associated with GAD on myopia. The x-axis is the beta value, which represents the risk of myopia among individuals with GAD. A beta <0 indicates that individuals with GAD is associated with more myopic MSE. Error bars represent 95% CIs. CI = confidence interval; GAD = generalized anxiety disorder; MR = Mendelian randomization; MSE = mean spherical equivalent; SNPs = single nucleotide polymorphisms.

Sensitivity Analyses

To guarantee the absence of directional pleiotropy, we performed the MR-Egger intercept test and MR pleiotropy residual sum and outlier (MR-PRESSO) test, with a P value >0.05 indicating the absence of horizontal pleiotropy.[38] We also applied Cochran Q using MR-Egger and IVW methods and funnel plots to evaluate heterogeneity, and a P value >0.05 coupled with symmetric distribution in the funnel plot represents no heterogeneity. Additionally, a leave-one-out sensitivity analysis was performed to test the effect of a single SNP on the causality. A reverse MR analysis was done with the same procedure to assess the potential reverse causality.

Results

Forward MR Analysis: Causal Effects of Anxiety on Myopia

We first conducted a forward-direction MR analysis to investigate the causal effect of anxiety on myopia. In the FinnGen Research, 5280 cases of 373 334 (1.41%) participants had GAD. Using IVW as the primary analysis, we found a significant relationship between GAD and MSE, showing individuals with GAD have a potentially higher risk of developing more myopic MSE (β = −0.138; 95% confidence interval [CI] −0.221 to −0.056, P = 0.001, Fig 2). Other methods show directionally consistent but no significant associations (MR-Egger regression, β = −0.350; 95% CI −0.677 to −0.023, P = 0.0655; weighted median, β = −0.065; 95% CI −0.175 to 0.044, P = 0.2418; weighted mode, β = −0.065; 95% CI −0.220 to 0.091, P = 0.4344, Fig 2 and 3B).

Sensitivity Analysis of the Forward Analysis

As shown in Table 1, the MR-Egger regression intercept analysis indicated no significant directional pleiotropy (intercept = 0.0252, P = 0.2239, Fig 3A), and similar results were observed with the MR-PRESSO global test (P = 0.3133). Additionally, no heterogeneity was found in the MR-Egger regression analysis (Q = 10.1643, P = 0.3374) and IVW method (Q = 12.0911, P = 0.279) via the Cochran Q test (Table 2). The MR-PRESSO test identified no outliers, and the leave-one-out sensitivity test validated the robustness of the findings (Fig 3C).

Reverse MR Analysis: Causal Effects of Myopia on Anxiety

We then performed a reverse MR analysis to investigate the causal effect of myopia on anxiety. All 4 MR methods show individuals with more myopic MSE may have higher risks

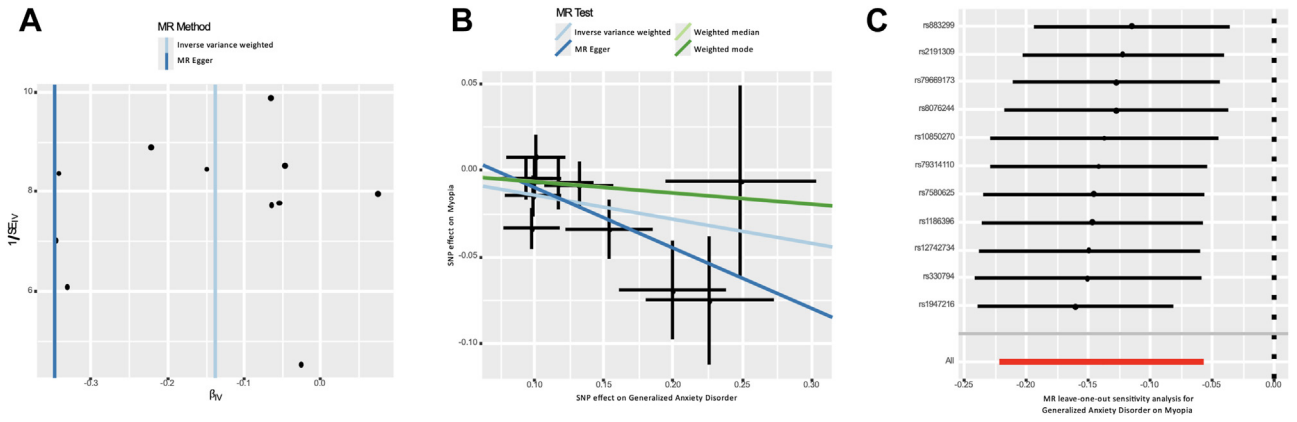


Figure 3. Analysis of causal effects of GAD on myopia. **A,** Funnel plot showing symmetric distribution, indicating no heterogeneity in the forward MR analysis. **B,** Scatter plot showing the correlation of genetic associations of GAD with myopia. **C,** Leave-one-out sensitivity analysis illustrating all statistical results are positioned on the left side of the vertical axis, suggesting good stability. GAD = generalized anxiety disorder; IV = instrumental variable; MR = Mendelian randomization; SE = standard error; SNPs = single nucleotide polymorphisms.

Table. 1. Results of Pleiotropy Tests

Exposure	Outcome	Number of SNPs	MR_Egger			PRESSO
			Intercept	*SE*	P *Value*	P Value
The forward MR analyses						
Generalized anxiety disorder	Myopia	11	0.0252	0.0193	0.2239	0.3133
The reverse MR analyses						
Myopia	Generalized anxiety disorder	253	−0.0016	0.0039	0.6774	0.115

MR = Mendelian randomization; PRESSO = pleiotropy residual sum and outlier; SE = standard error; SNPs = single nucleotide polymorphisms.
Results of pleiotropy tests in the forward MR analyses and the reverse MR analyses. No pleiotropy has been found in both forward and reverse MR analyses via the MR-Egger intercept test and the MR-PRESSO test.

of GAD but with no statistical significance (IVW: $\beta = -0.017$; 95% CI −0.051 to 0.161, $P = 0.3075$; MR-Egger: $\beta = -0.002$; 95% CI −0.084 to 0.081; weighted median method: $\beta = -0.018$; 95% CI −0.070 to 0.035; weighted mode method: $\beta = -0.006$; 95% CI −0.080 to 0.068, Fig 4).

Sensitivity Analyses of Reverse MR Analysis

As shown in Table 1, in the reverse 2-sample MR, both the MR-Egger regression (intercept = −0.0016, $P = 0.6774$, Fig S5, available at www.ophthalmologyscience.org) and the MR-PRESSO global test ($P = 0.115$) suggested an absence of directional pleiotropy. Furthermore, no evidence of heterogeneity was found in the MR-Egger regression intercept analysis and IVW method, as indicated by the Cochran Q test results: $Q = 278.3522$ ($P = 0.1134$) and $Q = 278.5446$ ($P = 0.1205$), respectively (Table 2, Fig S6, available at www.ophthalmologyscience.org). There was no outlier identified in the MR-PRESSO test, and the robustness of the results was verified by the leave-one-out sensitivity analysis (Fig S7, available at www.ophthalmologyscience.org).

Discussion

Our MR study found a 1-way causal relationship from GAD to myopia, showing that patients with GAD tend to have more myopic MSE (IVW, $\beta = -0.138$; 95% CI: −0.221 to −0.056; $P = 0.001$). However, there was no significant evidence that myopia increases the risk of GAD (IVW, $\beta = -0.017$; 95% CI: −0.051 to 0.161; $P = 0.3075$). Although the effect size was relatively small, the current MR analysis has confirmed a robust causal relationship between GAD and more myopic refractive error as validated by multiple sensitivity analyses.

Previous cross-sectional studies have consistently reported an association between anxiety and myopia. Nitzan et al utilized noncycloplegic monocular spherical equivalent measurement in Israeli adolescents and found that those with myopia had a twofold increased risk of anxiety and mood disorders.[39] Studies of Chinese university students based on noncycloplegic autorefraction also revealed a positive correlation between anxiety and myopia.[16] This study further established a causal relationship through MR, providing evidence that anxiety is a likely contributing factor to the development of myopia rather than a consequence of it. Though the effect size was relatively small ($\beta = -0.138$ diopters), it is biologically plausible since myopia is a multifactorial condition where individual genetic or environmental factors typically have small effects,[40,41] and the effect sizes reported in previous MR studies regarding myopia were mostly small (e.g., −0.27 diopters per 1 additional year of education; −0.02 diopters per 10 nmol/l increase in 25(OH)D; 0.04 diopters per standard deviation increase in birth weight).[42–44] Although modest, this does not undermine the validity or clarity of the causal relationship between GAD and myopia.

The causal relationship of GAD to a more myopic MSE may be attributed to the following mechanisms. Firstly, studies have suggested chronic stress-related hyperactivity in the sympathetic nervous system in patients with GAD, leading to the release of stress hormones such as catecholamine and

Table.2. Results of Heterogeneity Tests

Exposure	Outcome	MR-Egger			IVW		
		Q	*Q_df*	*Q_P Value*	*Q*	*Q_df*	*Q_P Value*
The forward MR analyses							
Generalized anxiety disorder	Myopia	10.1643	9	0.3374	12.0911	10	0.279
The reverse MR analyses							
Myopia	Generalized anxiety disorder	278.3522	251	0.1134	278.5446	252	0.1205

IVW = inverse variance weighted; MR = Mendelian randomization.
Results of heterogeneity tests for the forward MR analyses and reverse MR analyses. No heterogeneity has been found via MR-Egger and IVW methods.

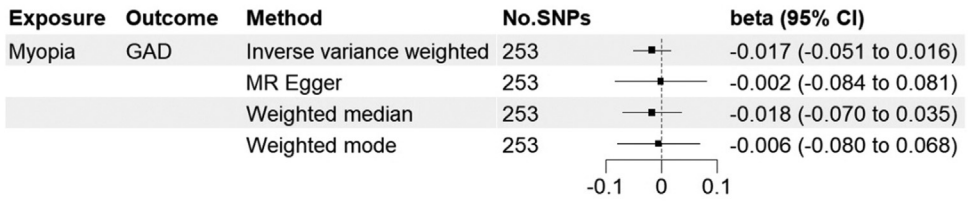


Figure 4. Forest plot showing the causal effects of SNPs associated with myopia on GAD. The x-axis is the beta value, which represents the risk of GAD associated with more myopic MSE. A beta <0 indicates that individuals with more myopic MSE are associated with a higher risk of GAD. Error bars represent 95% CIs, with the inclusion of "0" suggesting no statistical significance. CI = confidence interval; GAD = generalized anxiety disorder; MR = Mendelian randomization; MSE = mean spherical equivalent; SNPs = single nucleotide polymorphisms.

norepinephrine. These hormones would induce choroidal vasoconstriction, resulting in choroidal thinning and reduced blood flow, thus exacerbating scleral hypoxia, promoting extracellular matrix remodeling, and eventually leading to axial elongation.[20,45,46] Secondly, GAD may influence myopia via dysregulation of dopamine, a neurotransmitter known to play a protective role in the development and progression of myopia.[47] Hashikawa-Hobara et al observed a reduction in dopamine level in anxiety-like behaviors, mediated by calcitonin gene—related peptide.[48]

Besides the neuroendocrine mechanisms, another possible mechanism is that children with GAD may engage in less outdoor activity, thereby increasing their risk of myopia. Previous research has demonstrated that mental disorders (especially anxiety) were associated with less outdoor activity.[49] While reduced outdoor time was not a direct confounder between GAD and myopia, it could act as a potential mediator.[50]

Our study is the first to demonstrate a causal relationship between GAD and myopia using a comprehensive MR approach, highlighting the potential link between mental health and myopia risk. However, we do not advocate for mental health interventions solely for the purpose of myopia prevention. Instead, we emphasize the overall benefits of mental health education and interventions for children and adolescents, including improved emotional well-being, reduced stress, healthier lifestyle behaviors, and better academic outcomes. Given that school-aged children and adolescents are at a critical stage of both mental and physical development and are at the highest risk for myopia, integrating mental health education and supportive interventions in schools may promote not only visual and physical health but also overall well-being.

Several limitations should be noted. First, our study primarily utilizes GWAS data from European populations, limiting the generalizability of the findings to non-European groups who may have different genetic and environmental risk factors. Second, 2-sample MR, while useful for inferring causality, is limited in its ability to fully exclude reverse causality and the influence of confounders. Even though a reverse MR analysis was conducted, these risks persist. Third, one significant limitation is the reliance on genetic variants as IVs. The validity of MR depends heavily on the strength of the IVs, and though MR aims to reduce confounding, it cannot eliminate the risk of residual confounding and pleiotropy. Even with sensitivity tests like MR-PRESSO and MR-Egger, hidden biases may persist, potentially affecting the robustness of the results.[28,38] Lastly, the current MR analysis was based on the GWAS data from adults, which may limit the direct extrapolation to children. Nonetheless, the causal relationship between GAD and myopia is not likely to be affected since MR analysis relies on genetic variants as IVs, which are fixed at conception and generally stable across the lifespan. Future research is encouraged to provide more direct evidence regarding the effect of mental health interventions on myopia in children.

Our study demonstrates a significant causal relationship between GAD and a more myopic MSE, suggesting an important connection between mental health and eye health. Improving mental well-being can have overall benefits for children's health, including better vision.

Footnotes and Disclosures

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HUMAN SUBJECTS: No human subjects were included in this study. This secondary analysis used deidentified data sourced from public databases. The original studies obtained ethical approvals from institutional review boards. Informed consent was secured during the primary data collection phase. Because no new data collection or participant interaction took place, additional ethical approval was deemed unnecessary according to the Declaration of Helsinki and the guidelines for anonymized data reuse.

No animal subjects were used in this study.

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Conception and design: Wong, Qin, Liu, Zhang, Jiang, Wang, Liang, Han

Data collection: Wong, Qin, Liu, Zhang, Zeng, Han

Analysis and interpretation: Wong, Qin, Zhang, Han

Obtained funding: Han

Overall responsibility: Wong, Qin, Liu, Zhang, Jiang, Zeng, Wang, Liang, Han

Abbreviations and Acronyms:
CI = confidence interval; **GAD** = generalized anxiety disorder; **GWAS** = genome-wide association studies; **IV** = instrumental variable; **IVW** = inverse variance weighted; **MR** = Mendelian randomization; **MSE** = mean spherical equivalent; **PRESSO** = pleiotropy residual sum and outlier; **SNP** = single nucleotide polymorphism.

Keywords:
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References

1. Holden BA, Fricke TR, Wilson DA, et al. Global prevalence of myopia and high myopia and temporal trends from 2000 through 2050. *Ophthalmology*. 2016;123:1036–1042.
2. Williams KM, Bertelsen G, Cumberland P, et al. Increasing prevalence of myopia in Europe and the impact of education. *Ophthalmology*. 2015;122:1489–1497.
3. Williams KM, Verhoeven VJM, Cumberland P, et al. Prevalence of refractive error in Europe: the European Eye Epidemiology (E3) Consortium. *Eur J Epidemiol*. 2015;30:305–315.
4. Németh J, Tapasztó B, Aclimandos WA, et al. Update and guidance on management of myopia. European Society of Ophthalmology in cooperation with International Myopia Institute. *Eur J Ophthalmol*. 2021;31:853–883. https://doi.org/10.1177/1120672121998960.
5. Haarman AEG, Enthoven CA, Tideman JWL, et al. The complications of myopia: a review and meta-analysis. *Invest Ophthalmol Vis Sci*. 2020;61:49.
6. Landreneau JR, Hesemann NP, Cardonell MA. Review on the myopia pandemic: epidemiology, risk factors, and prevention. *Mo Med*. 2021;118:156–163.
7. Biswas S, El Kareh A, Qureshi M, et al. The influence of the environment and lifestyle on myopia. *J Physiol Anthropol*. 2024;43:7.
8. Dutheil F, Oueslati T, Delamarre L, et al. Myopia and near work: a systematic review and meta-analysis. *Int J Environ Res Public Health*. 2023;20:875.
9. Morgan IG, Wu PC, Ostrin LA, et al. IMI risk factors for myopia. *Invest Ophthalmol Vis Sci*. 2021;62:3.
10. Foreman J, Salim AT, Praveen A, et al. Association between digital smart device use and myopia: a systematic review and meta-analysis. *Lancet Digit Health*. 2021;3:e806–e818.
11. Wu Y, Li X, Ji X, et al. Trends in the epidemiology of anxiety disorders from 1990 to 2021: a global, regional, and national analysis with a focus on the sociodemographic index. *J Affect Disord*. 2025;373:166–174.
12. Yang X, Fang Y, Chen H, et al. Global, regional and national burden of anxiety disorders from 1990 to 2019: results from the global burden of disease study 2019. *Epidemiol Psychiatr Sci*. 2021;30:e36.
13. Sacco R, Camilleri N, Eberhardt J, et al. A systematic review and meta-analysis on the prevalence of mental disorders among children and adolescents in Europe. *Eur Child Adolesc Psychiatry*. 2024;33:2877–2894.
14. Nitzan I, Shmueli O, Safir M. Association of myopia with anxiety and mood disorders in adolescents. *Eye Lond Engl*. 2024;33:2877–2894.
15. Li Z, Wei J, Lu S, et al. Association between myopia and anxiety: a cross-sectional study based on Chinese university freshmen. *Ann Transl Med*. 2023;11:298.
16. Zhang H, Gao H, Zhu Y, et al. Relationship between myopia and other risk factors with anxiety and depression among Chinese university freshmen during the COVID-19 pandemic. *Front Public Health*. 2021;9:774237.
17. Zhu X, Meng J, Han C, et al. CCL2-mediated inflammatory pathogenesis underlies high myopia-related anxiety. *Cell Discov*. 2023;9:94.
18. Li D, Chan VF, Virgili G, et al. Impact of vision impairment and ocular morbidity and their treatment on depression and anxiety in children: a systematic review. *Ophthalmology*. 2022;129:1152–1170.
19. Li Q, Yang J, He Y, et al. Investigation of the psychological health of first-year high school students with myopia in guangzhou. *Brain Behav*. 2020;10:e01594.
20. Lin S, Zhu B, Wang T, et al. Sympathetic nervous system activity is associated with choroidal thickness and axial length in school-aged children. *Br J Ophthalmol*. 2024;108:405–410.
21. Bandelow B, Michaelis S. Epidemiology of anxiety disorders in the 21st century. *Dialogues Clin Neurosci*. 2015;17:327.
22. Smith GD, Ebrahim S. "mendelian randomization": can genetic epidemiology contribute to understanding environmental determinants of disease? *Int J Epidemiol*. 2003;32:1–22.
23. Cai XB, Shen SR, Chen DF, et al. An overview of myopia genetics. *Exp Eye Res*. 2019;188:107778.
24. Tedja MS, Haarman AEG, Meester-Smoor MA, et al. IMI – myopia genetics report. *Invest Ophthalmol Vis Sci*. 2019;60:M89–M105.
25. Hysi PG, Wojciechowski R, Rahi JS, Cj H. Genome-wide association studies of refractive error and myopia, lessons learned, and implications for the future. *Invest Ophthalmol Vis Sci*. 2014;55:3344–3351.
26. Reading mendelian randomisation studies: a guide, glossary, and checklist for clinicians - PubMed. https://pubmed.ncbi.nlm.nih.gov/30002074/. Accessed September 27, 2024.

27. Bowden J, Holmes MV. Meta-analysis and mendelian randomization: a review. *Res Synth Methods*. 2019;10:486−496.
28. Birney E. Mendelian randomization. *Cold Spring Harb Perspect Med*. 2022;12:a041302.
29. Sekula P, Del Greco MF, Pattaro C, Köttgen A. Mendelian randomization as an approach to assess causality using observational data. *J Am Soc Nephrol JASN*. 2016;27:3253−3265.
30. Davies NM, Holmes MV, Davey Smith G. Reading mendelian randomisation studies: a guide, glossary, and checklist for clinicians. *BMJ*. 2018;362:k601.
31. Ghorbani Mojarrad N, Plotnikov D, Williams C, Guggenheim JA; for the UK Biobank Eye and Vision Consortium. Association between polygenic risk score and risk of myopia. *JAMA Ophthalmol*. 2020;138:7.
32. Fotouhi A, Morgan IG, Iribarren R, et al. Validity of noncycloplegic refraction in the assessment of refractive errors: the Tehran eye study. *Acta Ophthalmol (Copenh)*. 2012;90:380−386.
33. Sanfilippo PG, Chu B, Bigault O, et al. What is the appropriate age cut-off for cycloplegia in refraction? *Acta Ophthalmol (Copenh)*. 2014;92:e458−e462.
34. Krantz EM. Measuring refraction in adults in epidemiological studies. *Arch Ophthalmol*. 2010;128:88.
35. Ma Z, Zhao M, Zhao H, Qu N. Causal role of immune cells in generalized anxiety disorder: mendelian randomization study. *Front Immunol*. 2024;14:1338083.
36. Reay WR, Cairns MJ. Advancing the use of genome-wide association studies for drug repurposing. *Nat Rev Genet*. 2021;22:658−671.
37. Guidelines for performing mendelian randomization investigations: update for summer 2023 - PMC. https://pmc.ncbi.nlm.nih.gov/articles/PMC7384151/. Accessed October 30, 2024.
38. Verbanck M, Chen CY, Neale B, Do R. Detection of widespread horizontal pleiotropy in causal relationships inferred from Mendelian randomization between complex traits and diseases. *Nat Genet*. 2018;50:693−698.
39. Nitzan I, Shmueli O, Safir M. Association of myopia with anxiety and mood disorders in adolescents. *Eye*. 2024;38:3016−3018.
40. The Consortium for Refractive Error and Myopia, The UK Eye and Vision Consortium, 23andMe Inc.. Meta-analysis of 542,934 subjects of european ancestry identifies new genes and mechanisms predisposing to refractive error and myopia. *Nat Genet*. 2020;52:401−407.
41. Lee SSY, Stapleton F, MacGregor S, Mackey DA. Genome-wide association studies, polygenic risk scores and mendelian randomisation: an overview of common genetic epidemiology methods for ophthalmic clinicians. *Br J Ophthalmol*. 2025;109:433−441.
42. Mountjoy E, Davies NM, Plotnikov D, et al. Education and myopia: assessing the direction of causality by mendelian randomisation. *BMJ*. 2018;361:k2022.
43. Cuellar-Partida G, Williams KM, Yazar S, et al. Genetically low vitamin D concentrations and myopic refractive error: a mendelian randomization study. *Int J Epidemiol*. 2017;46:1882−1890.
44. Plotnikov D, Williams C, Guggenheim JA. Association between birth weight and refractive error in adulthood: a mendelian randomisation study. *Br J Ophthalmol*. 2020;104:214−219.
45. Hughes JW, Watkins L, Blumenthal JA, et al. Depression and anxiety symptoms are related to increased 24-hour urinary norepinephrine excretion among healthy middle-aged women. *J Psychosom Res*. 2004;57:353−358.
46. Kallen VL, Tulen JHM, Utens EMWJ, et al. Associations between HPA axis functioning and level of anxiety in children and adolescents with an anxiety disorder. *Depress Anxiety*. 2008;25:131−141.
47. Zhou X, Pardue MT, Iuvone PM, Qu J. Dopamine signaling and myopia development: what are the key challenges. *Prog Retin Eye Res*. 2017;61:60−71.
48. Hashikawa-Hobara N, Fujiwara K, Hashikawa N. CGRP causes anxiety via HP1γ-KLF11-MAOB pathway and dopamine in the dorsal hippocampus. *Commun Biol*. 2024;7:322.
49. Hiles SA, Lamers F, Milaneschi Y, Penninx BWJH. Sit, step, sweat: longitudinal associations between physical activity patterns, anxiety and depression. *Psychol Med*. 2017;47:1466−1477.
50. He X, Sankaridurg P, Wang J, et al. Time outdoors in reducing myopia. *Ophthalmology*. 2022;129:1245−1254.